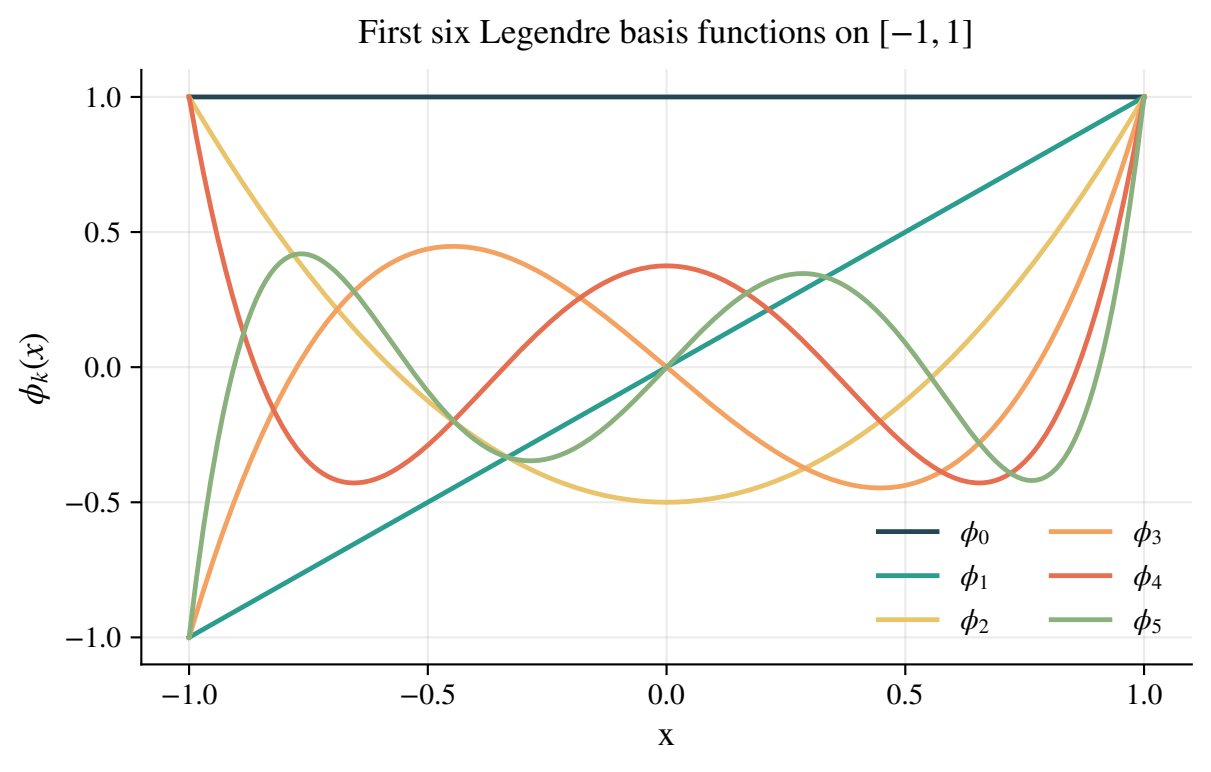
First six Legendre basis functions on [−1, 1]
1.0
0.5
0.0
−0.5
−1.0
$\phi_k(x)$
−1.0
−0.5
0.0
0.5
1.0
x
ϕ_0
ϕ_1
ϕ_2
ϕ_3
ϕ_4
ϕ_5